CUMULATIVE HIERARCHIES OF UNIVERSES AND THEIR EQUIVALENCE IN TYPE THEORY

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OUTLINE

1. Introduction

INTRODUCTION : UNIVERSES

$$\boxed{A, B, a, b ::= \Pi\, A\, B \mid \mathsf{U} \mid \lambda_{x:A}\, b \mid c\, a}$$

- The type of all types U leads to *Girard's paradox*
- We need a universe hierarchy $(\mathsf{U}_i)_{i \in \mathbb{N}}$
- We can ask this hierarchy to be cumulative:

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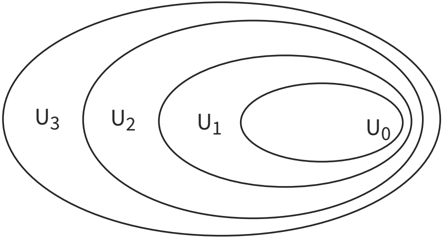

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Two conflicting worlds :

- Practitioners of type theory want *implicit* universes (*Russell*-style).
- Type theorists use *explicit* universes for semantics and meta-theory (*Tarski*-style).
- We need an equivalence result
  - $\Rightarrow$ Legitimizing this informal practice
  - $\Rightarrow$ Rigorous interpretation in the usual models of type theory
  - $\Rightarrow$ Extension of initiality to implicit syntax
- Assaf 14, Thiré 20, Felicissimo and Winterhalter 24: Equivalence proof for the calculus of construction, with β as a reduction but doesn't adapt to $\beta\eta$ judgemental conversion.

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- Syntactic equivalence of the Russell and Tarski presentations for cumulative universes
  - ⇒ The proof is formalised in Rocq !
    https://github.com/raphael-sterbac/Russell-Tarski-Equivalence
- Relate Generalised Algebraic Theory (GAT) and Pure Type System (PTS) presentations, with typed judgemental $\beta\eta$ conversion.

To do this:

- Key lemmas : injectivity of type constructors and "no-confusion" (only for the GAT).
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GAT ($\mathcal{T}_G$) | Tarski ($\mathcal{T}_T$) | Russell ($\mathcal{T}_R$) | PTS ($\mathcal{T}_P$)

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TARSKI UNIVERSES : $\mathcal{T}_T$

- We define the theory $\mathcal{T}_T$ with Universes **à la Tarski** (Martin Löf 84)

→ A *fully explicit* presentation

$$A, B ::= \Pi\ A\ B \mid \mathsf{U}_n \mid \mathsf{El}_n(a)$$

$$a, b ::= \lambda(A, B, b) \mid \mathsf{app}(A, B, c, a) \mid \Pi^n\ a\ b \mid \mathsf{U}_n^m \mid \uparrow_n^m (a)$$

- With the following equations:

$$\mathsf{El}_m(\mathsf{U}_l^m) = \mathsf{U}_l \qquad \uparrow_m^n \uparrow_l^m (a) = \uparrow_l^n (a)$$

$$\mathsf{El}_l(\Pi^l(a, b)) = \Pi(\mathsf{El}_l(a), \mathsf{El}_l(b)) \qquad \uparrow_m^n (\mathsf{U}_l^m) = \mathsf{U}_l^n$$

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→ We can present Type theory with Tarski universes as a GAT $\mathcal{T}_G$

- This theory $\mathcal{T}_G$ has an *explicit* substitution calculus, we have a forgetful map proj to $\mathcal{T}_T$

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→ Types and terms are conflated

$$\mathsf{U}_0 : \mathsf{U}_1, \quad \mathsf{U}_1 : \mathsf{U}_2, \quad \dots \quad \mathsf{U}_n : \mathsf{U}_{n+1}, \quad \dots$$

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- $|\mathsf{U}_n| = \mathsf{U}_n$.
- $|\mathsf{U}^n_m| = \mathsf{U}_m$.
- $|\mathsf{app}(A, B, c, a)| = \mathsf{app}(|A|, |B|, |c|, |a|)$.
- $|\lambda(A, B, b)| = \lambda(|A|, |B|, |b|)$.
- $|v_i| = v_i$.
- $|\mathsf{El}_n(a)| = |a|$.
- $|\uparrow^n_m (a)| = |a|$.
- $|\Pi^n\, a\, b| = \Pi(|a|, |b|)$.
- $|\Pi(A, B)| = \Pi(|A|, |B|)$.

$$\begin{tikzcd} \boxed{\text{GAT } (\mathcal{T}_G)} \arrow[r, "\text{proj}"] & \boxed{\text{Tarski } (\mathcal{T}_T)} \arrow[r, "|.|"] & \boxed{\text{Russell } (\mathcal{T}_R)} \end{tikzcd}$$

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PURE TYPE SYSTEM : PRESENTATION WITHOUT ANNOTATIONS

- The final theory $\mathcal{T}_P$: unannotated application and abstraction

$\rightarrow$ Presentation as a "Pure Type System"

$$\begin{array}{ccc} \lambda(A, B, b) \mid \mathrm{app}(A, B, c, a) & \overset{\text{strip}}{\Longrightarrow} & \lambda_{x:A}\, b \mid c\ a \\ \text{(System } \mathcal{T}_R\text{)} & & \text{(System } \mathcal{T}_P\text{)} \end{array}$$

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SUMMARY

Those four theories have forgetful / erasure maps:

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Goal: find inverses → "any judgement can be lifted back in a unique way".

3. Equivalence result

EQUIVALENCE BETWEEN $\mathcal{T}_T$ AND $\mathcal{T}_R$: RUSSELL AND TARSKI

We prove $\mathcal{T}_T \Leftrightarrow \mathcal{T}_R$, relating *Russell* and *Tarski* universes.

We could try to prove the following lemma :

> If $\Gamma \vdash u_0 : A_0$ and $\Gamma \vdash u_1 : A_1$ in $\mathcal{T}_T$ with $|u_0| = |u_1|$
> Then, $\Gamma \vdash A_0 = A_1$ and $\Gamma \vdash u_0 = u_1 : A_0$.

"If two terms are syntactically the same in $\mathcal{T}_R$, then they should have the same behaviour in $\mathcal{T}_T$ and thus be convertible"

$\rightarrow$ Or not... take $x : \mathrm{U}_n$ and $\uparrow_n^m x : \mathrm{U}_m$
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EQUIVALENCE BETWEEN $\mathcal{T}_T$ AND $\mathcal{T}_R$: UNIQUENESS LEMMA

Deal with the case of codes $u_0 : \mathsf{U}_{n_0}$ and $u_1 : \mathsf{U}_{n_1}$ having the same erasure $|u_0| = |u_1|$.

→ Find two convertible codes $v_0 = v_1$ in a lower universe U_k that lifts to u_0 and u_1

⇒ Two terms in $\mathcal{T}_T$ with the same erasure are convertible *modulo lifting* !

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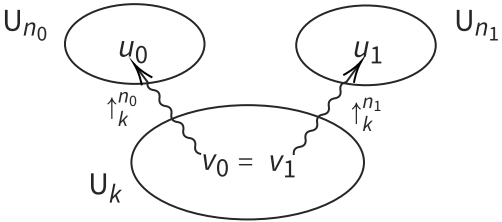

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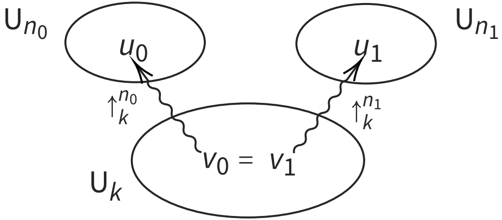

⇒ Two terms in $\mathcal{T}_T$ with the same erasure are convertible *modulo lifting* !

EQUIVALENCE BETWEEN $\mathcal{T}_T$ AND $\mathcal{T}_R$: UNIQUENESS LEMMA

- If $\Gamma \vdash A$ type and $\Gamma \vdash B$ type in $\mathcal{T}_T$, with $|A| = |B|$, then $\Gamma \vdash A = B$.
- If $\Gamma \vdash u_0 : A_0$ and $\Gamma \vdash u_1 : A_1$ in $\mathcal{T}_T$ with $|u_0| = |u_1|$ then :
  - *1st case* : $\Gamma \vdash A_0 = A_1$ and $\Gamma \vdash u_0 = u_1 : A_0$.
  - *2nd case* : $\Gamma \vdash A_0 = \mathrm{U}_{n_0}$ and $\Gamma \vdash A_1 = \mathrm{U}_{n_1}$, with $\Gamma \vdash u_0 = \uparrow_k^{n_0} (v_0) : A_0$ and $\Gamma \vdash u_1 = \uparrow_k^{n_1} (v_1) : A_1$ such that $\Gamma \vdash v_0 = v_1 : \mathrm{U}_k$.

- Proof by mutual induction on the size of A and u_0.
- Injectivity of $\mathrm{U}_\bullet$ and no-confusion between $\mathrm{U}_\bullet$ and $\Pi(-,-)$ are necessary.

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With this, we show that we can uniquely lift judgements from $\mathcal{T}_R$ to $\mathcal{T}_T$, by exhibiting a translation from $\mathcal{T}_R$ to $\mathcal{T}_T$.

If $\Gamma \vdash J$ in $\mathcal{T}_R$ then there exists a unique judgment $\Gamma_T \vdash J_T$ modulo conversion in $\mathcal{T}_T$ such that $|\Gamma_T| = \Gamma$ and $|J_T| = J$.

$\Rightarrow$ This proves the equivalence of Tarski and Russell universes.

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- The proof technique extends to $\mathcal{T}_R \Leftrightarrow \mathcal{T}_P$, adapting the result of Streicher's book

→ We get the following equivalences

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EXTENDING THE RESULT

This proof is elementary, and can be done in primitive recursive arithmetic.

It adapts to various additions to the type theory.

- Addition of Data Types, for instance $\mathrm{Id}_A(a, b)$, List, Nat...
- Internal level *judgements* and universe polymorhphism (Bezem, Coquand, Dybjer and Escardo 2024), see next talk!

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Contributions and results :

- Equivalence between Tarski-style and Russell-style universes by a modular proof method
- Equivalence between GAT and PTS presentations of Type Theory

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Thank you for your attention.